

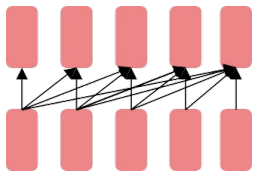


ECE/CS/ME 539
Introduction to Artificial Neural Networks

Advanced Topics: Modern Language Models

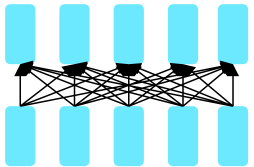


Types of common architectures for NLP



Decoder-only

- Language models! What we've seen so far.
- Nice to generate from; can't condition on future words
- **Examples:** GPT-2, GPT-3, LaMDA



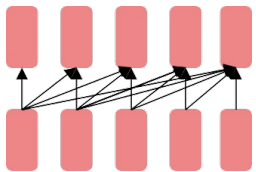
Encoder-only

- Gets bidirectional context – can condition on future!
- Great for text analysis.
- **Examples:** BERT and its many variants, e.g. RoBERTa

The neural architecture influences how to train, and what they can be useful for.

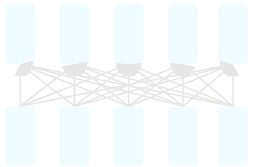


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Pretraining through language modeling [\[Dai and Le, 2015\]](#)

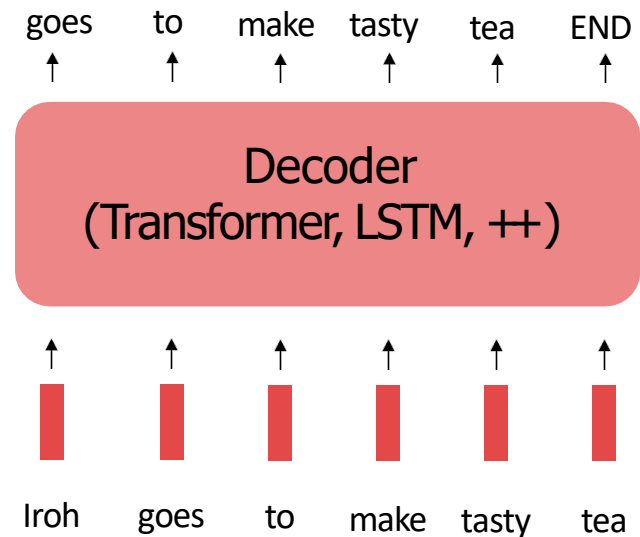
One common/effective way to train decoder-only models is by language modeling.

Language modeling

Model the probability distribution over words given their past contexts $p_{\theta}(w_t | w_{1:t-1})$.

Pretraining through language modeling:

Train a neural network to perform language modeling on a large amount of text.



Generative Pretrained Transformer (GPT) [[Radford et al., 2018](#)]

Objective: Language modeling

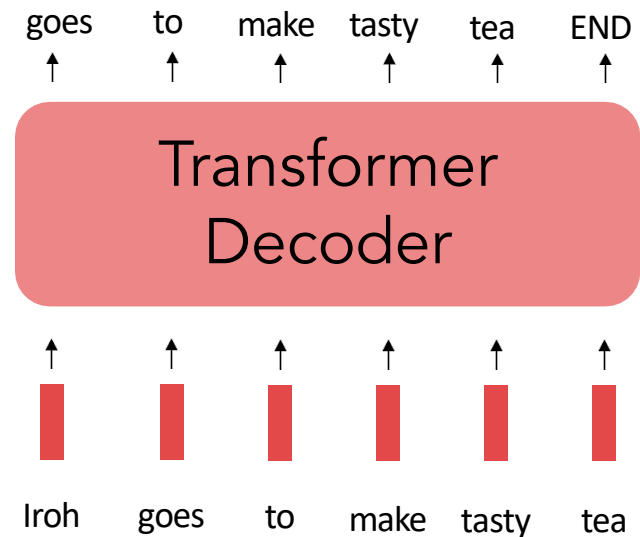
Model is trained to predict the next word, anywhere in a sentence.

Model

- 12 multi-headed self-attention layers with causal masking.
- Byte-pair encoding

Training data:

- BooksCorpus: over 7000 books



Generative Pretrained Transformer (GPT) [[Radford et al., 2018](#)]

GPT results on various *natural language inference* datasets.

Method	MNLI-m	MNLI-mm	SNLI	SciTail	QNLI	RTE
ESIM + ELMo [44] (5x)	-	-	<u>89.3</u>	-	-	-
CAFE [58] (5x)	80.2	79.0	<u>89.3</u>	-	-	-
Stochastic Answer Network [35] (3x)	<u>80.6</u>	<u>80.1</u>	-	-	-	-
CAFE [58]	78.7	77.9	88.5	<u>83.3</u>		
GenSen [64]	71.4	71.3	-	-	<u>82.3</u>	59.2
Multi-task BiLSTM + Attn [64]	72.2	72.1	-	-	82.1	61.7
Finetuned Transformer LM (ours)	82.1	81.4	89.9	88.3	88.1	56.0



Increasingly convincing generations (GPT2) [[Radford et al., 2018](#)]

We mentioned how pretrained decoders can be used **in their capacities as language models**.

GPT-2, a larger version of GPT trained on more data, was shown to produce relatively convincing samples of natural language.

Context (human-written): In a shocking finding, scientist discovered a herd of unicorns living in a remote, previously unexplored valley, in the Andes Mountains. Even more surprising to the researchers was the fact that the unicorns spoke perfect English.

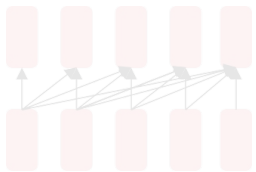
GPT-2: The scientist named the population, after their distinctive horn, Ovid's Unicorn. These four-horned, silver-white unicorns were previously unknown to science.

Now, after almost two centuries, the mystery of what sparked this odd phenomenon is finally solved.

Dr. Jorge Pérez, an evolutionary biologist from the University of La Paz, and several companions, were exploring the Andes Mountains when they found a small valley, with no other animals or humans. Pérez noticed that the valley had what appeared to be a natural fountain, surrounded by two peaks of rock and silver snow.

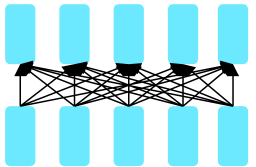


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Encoder-only

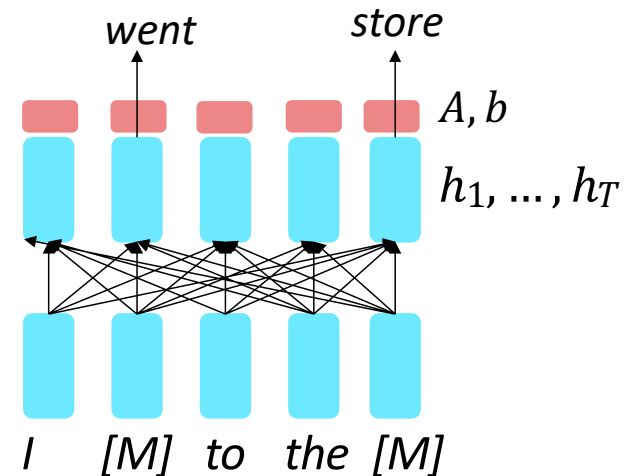
- Gets bidirectional context – can condition on future!
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Pretraining encoder-only models

Idea: Replace a fraction of words in the input with a special [MASK] token; predict these words.

- Only add loss terms from words that are “masked out.”
- Called **Masked LM**.



BERT: Bidirectional Encoder Representations from Transformers

Objective 1: Masked language modeling.

15% of tokens predicted in each sentence.

- Replaces word with [MASK] 80% of the time.
- Replaces word with a random token 10% of the time.
- Leaves word unchanged 10% of the time.

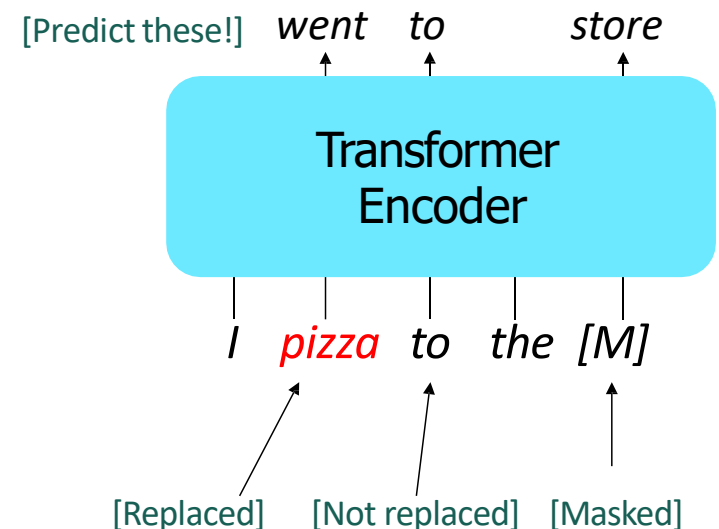
Objective 2: Next sentence verification.

- Input: <CLS> + Sentence A + Sentence B
- Logistic regression applied to <CLS> token to predict if sentence B follows sentence A in original text.

Trained on:

- BooksCorpus (800 million words)
- English Wikipedia (2,500 million words)

[\[Devlin et al., 2018\]](#)



BERT



BERT: Bidirectional Encoder Representations from Transformers

Finetuning BERT led to new state-of-the-art results on a broad range of tasks.

System	MNLI-(m/mm) 392k	QQP 363k	QNLI 108k	SST-2 67k	CoLA 8.5k	STS-B 5.7k	MRPC 3.5k	RTE 2.5k	Average -
Pre-OpenAI SOTA	80.6/80.1	66.1	82.3	93.2	35.0	81.0	86.0	61.7	74.0
BiLSTM+ELMo+Attn	76.4/76.1	64.8	79.8	90.4	36.0	73.3	84.9	56.8	71.0
OpenAI GPT	82.1/81.4	70.3	87.4	91.3	45.4	80.0	82.3	56.0	75.1
BERT _{BASE}	84.6/83.4	71.2	90.5	93.5	52.1	85.8	88.9	66.4	79.6
BERT _{LARGE}	86.7/85.9	72.1	92.7	94.9	60.5	86.5	89.3	70.1	82.1

[[Devlin et al., 2018](#)]

QQP: Quora Question Pairs (detect paraphrase questions)

QNLI: Natural language inference over question answering data

SST-2: sentiment analysis

MRPC: paraphrasing

RTE: natural language inference

CoLA: linguistic acceptability (detect whether sentences are grammatical.)

STS-B: semantic textual similarity



BERT: Bidirectional Encoder Representations from Transformers

Problem: Auto-regressive generation (word-by-word) is hard, due to future conditioning. Under-performs on generation tasks.

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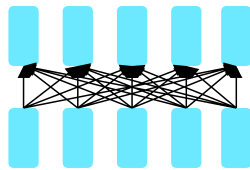
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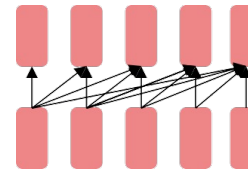
STS-B: semantic textual similarity



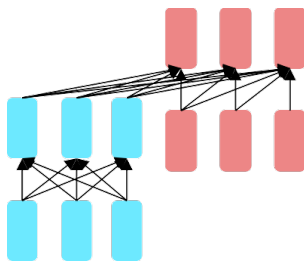
Types of common architectures for NLP



Encoder-only



Decoder-only



How about encoder-decoders?

Examples: Transformer, T5, Meena

Language Modeling is a common choice.

- The encoder benefits from bidirectional context;
- The decoder is used to train the whole model through language modeling.

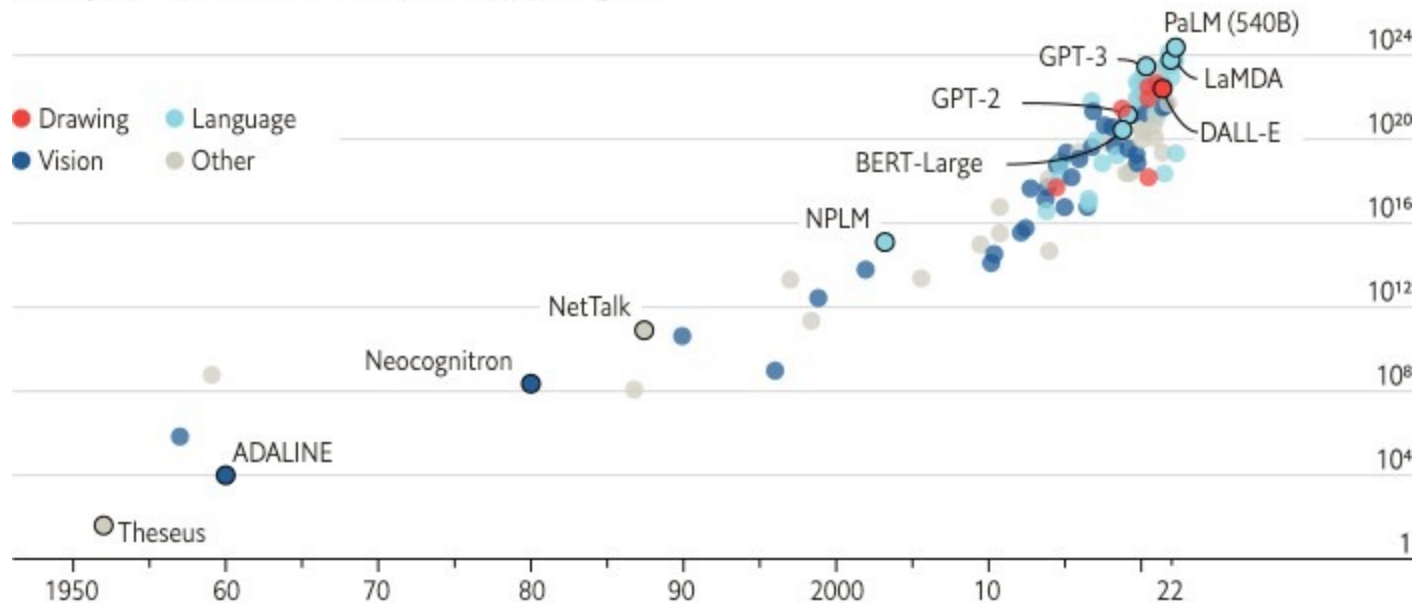


The Value of Scale

The blessings of scale

AI training runs, estimated computing resources used

Floating-point operations, selected systems, by type, log scale

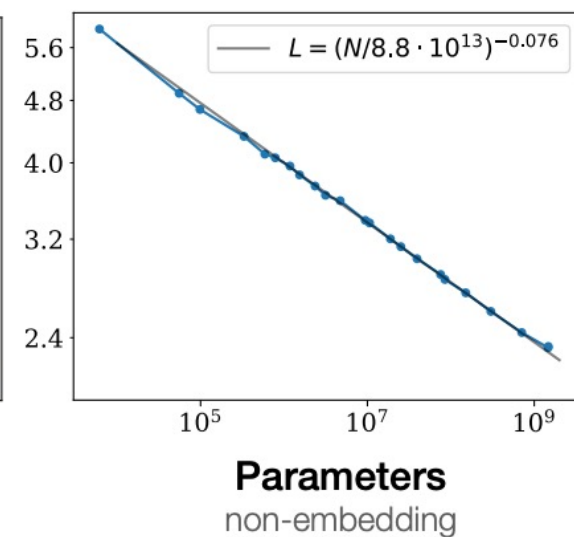
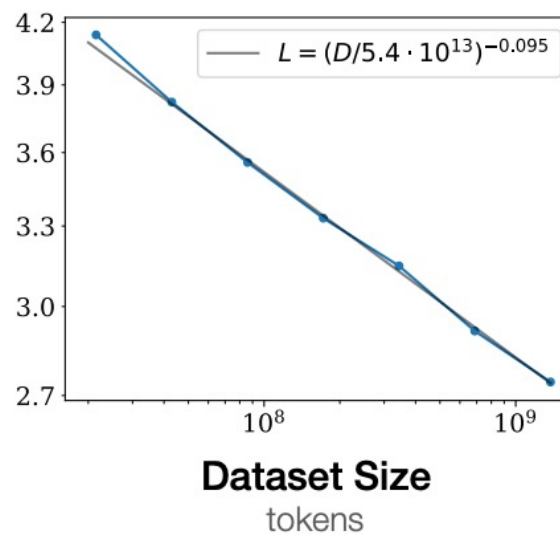
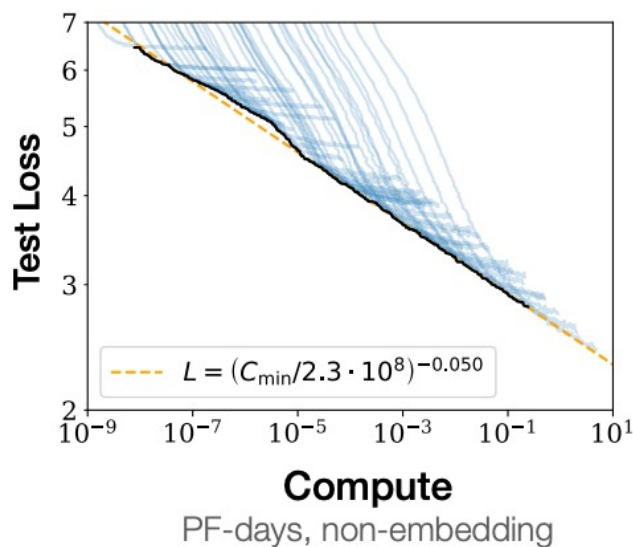


Sources: "Compute trends across three eras of machine learning", by J. Sevilla et al., arXiv, 2022; Our World in Data

<https://www.economist.com/interactive/briefing/2022/06/11/huge-foundation-models-are-turbo-charging-ai-progress>



Scaling Properties of GPT-3



How to make modern Language Models useful?

...beyond finetuning on your task.

Zero-Shot and
Few-Shot Prompting

Reinforcement Learning from
Human Feedback (RLHF)



Zero-Shot Prompting

We can get interesting behavior from Language Models simply by prompting the model in appropriate ways that clearly specify the task.

No task specific finetuning required.

Task: Summarization

Prompt: <ARTICLE CONTENT> TL;DR: ____

ROUGE

	R-1	R-2	R-L
Bottom-Up Sum	41.22	18.68	38.34
Lede-3	40.38	17.66	36.62
Seq2Seq + Attn	31.33	11.81	28.83
GPT-2 TL;DR:	29.34	8.27	26.58
Random-3	28.78	8.63	25.52

[[Radford et al., 2019](#)]

Task: Content Retrieval

Prompt: Passage: <PASSAGE> Q:
<QUESTION> A: ____

	LAMBADA (PPL)	LAMBADA (ACC)	CBT-CN (ACC)
SOTA	99.8	59.23	85.7
117M	35.13	45.99	87.65
345M	15.60	55.48	92.35
762M	10.87	60.12	93.45
1542M	8.63	63.24	93.30

[[Radford et al., 2019](#)]

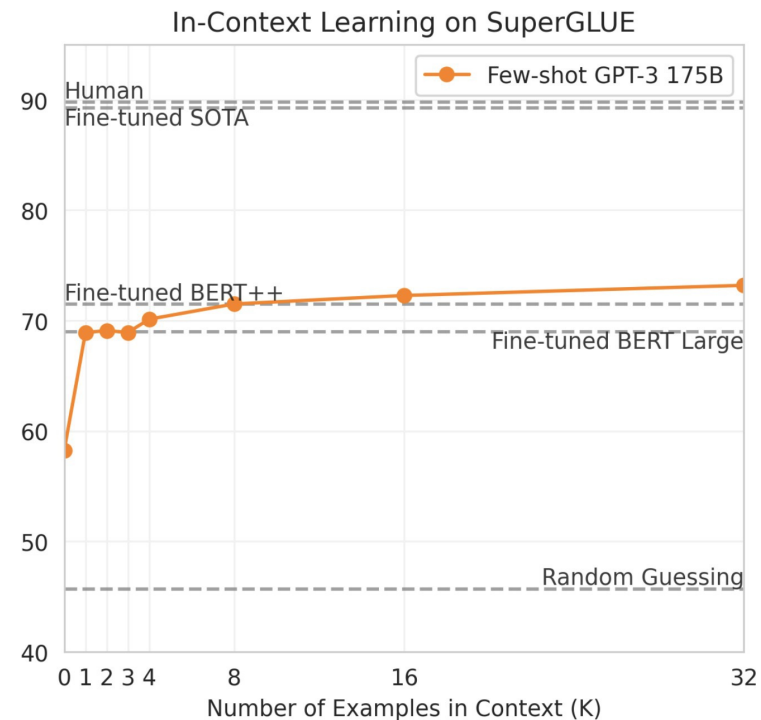


Few-Shot Prompting

Idea: Specify the task by prepending correctly answered examples before the one you want an answer for.

1 thanks => merci
2 hello => bonjour
3 mint => menthe
4 wall => mur
5 otter => loutre
6 bread => _____

In-context learning



[Brown et al., 2020]



Limits of Prompting/Language Models

Some tasks seem too hard for even large LMs to learn through prompting alone. Especially tasks involving richer, multi-step reasoning.

Physical Reasoning



What happens if I hold a piece of paper horizontally with two hands and I let go of one hand?



If you were holding a piece of paper horizontally with two hands and let go of one hand, the paper would likely tilt or rotate in the direction of the hand that you let go of. This is because the paper is no longer being held evenly on both sides, and the weight distribution is now uneven. If the paper is a light weight, it will drift to the side where you let go of and fall to the ground. If the paper has a little more weight or if you have a strong grip, it will stay in air for a moment before falling down.

Failure of ChatGPT in physical reasoning (example by Yann Lecun).

Math

$$19583 + 29534 = 49117$$

$$98394 + 49384 = 147778$$

$$29382 + 12347 = 41729$$

$$93847 + 39299 = ?$$

Factual Errors

Spatial/Temporal Reasoning

Bias/Discrimination



How to make modern Language Models useful?

...beyond finetuning on your task.

Zero-Shot and Few-Shot Prompting

- + No finetuning needed, prompt engineering (e.g. CoT) can improve performance
- Can be hard to find the right prompt.
- Behavior largely dictated by pretraining corpus

Reinforcement Learning from Human Feedback (RLHF)



RLHF: Optimizing for human preferences

Key contribution to: GPT-3.5 → Chat-GPT

Suppose for each generated output, we had a way to obtain a human reward reflecting how good the model's generation was.

Prompt

SAN FRANCISCO,
California (CNN) -- A
magnitude 4.2 earthquake
shook the San Francisco
...
overturn unstable
objects. TL;DR: _____

Answer 1:

An earthquake hit
San Francisco.
There was minor
property damage,
but no injuries.

$$R(s_1) = 8.0$$

Answer 2:

The Bay Area has good
weather but is prone
to earthquakes and
wildfires.

$$R(s_2) = 1.0$$

Idea: Finetune the model to maximize expected reward of generated samples.

$$\max_{\theta} E_{s \sim p_{\theta}(x)}[R(s)]$$



Reinforcement Learning to the rescue

- The field of reinforcement learning (RL) has studied these (and related) problems for many years now [Sutton and Barto, 1998]
- RLHF based on a technique called **policy gradient / REINFORCE**
- If you want to know how policy gradients work, try ECE/CS760.



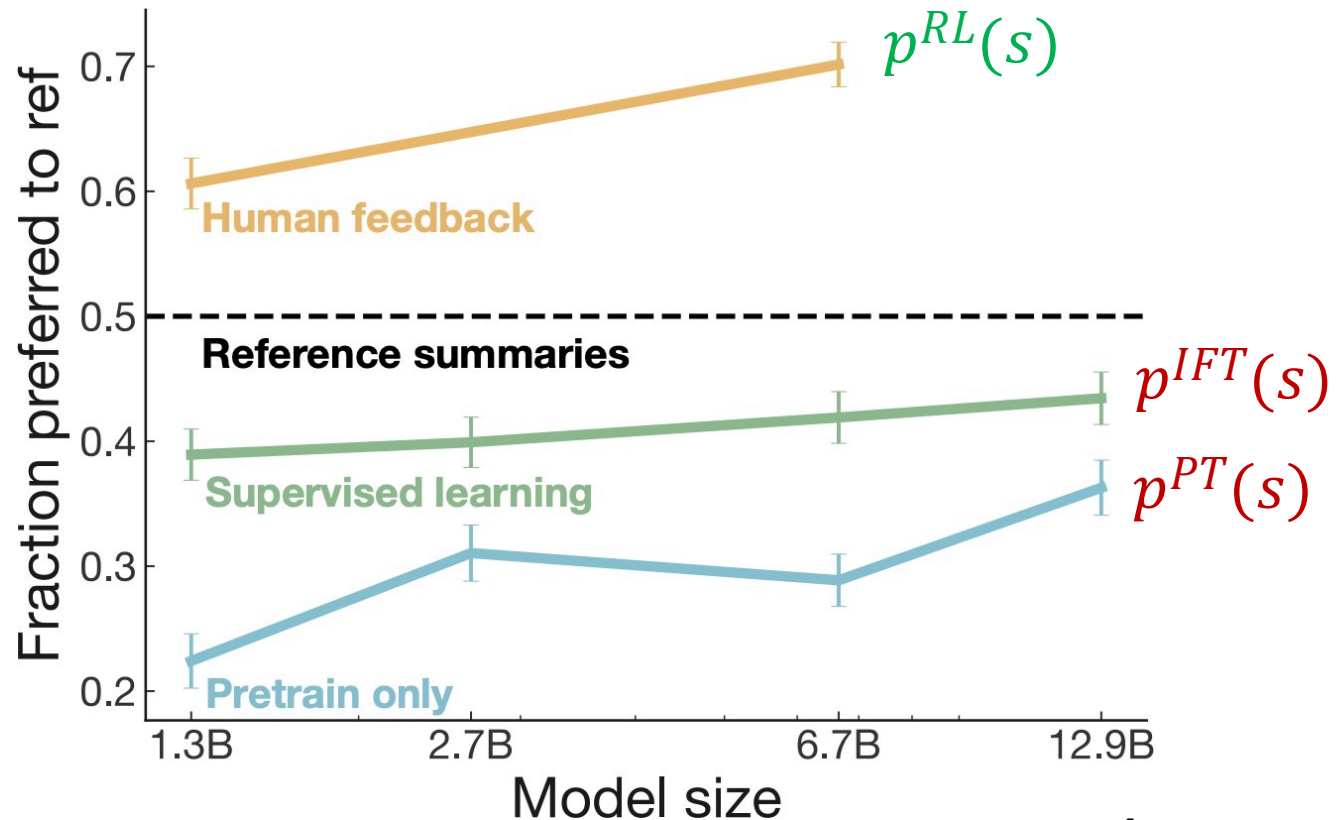
Reinforcement Learning

An Introduction
second edition

Richard S. Sutton and Andrew G. Barto



RLHF when scaled to thousands of tasks provides gains over pretraining + finetuning.



[Stiennon et al., 2020]



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...beyond finetuning on your task.

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Reinforcement Learning from Human Feedback (RLHF)

- + Directly model preferences (cf. language modeling), generalize beyond labeled data
- + More effective than prompting and finetuning
- RL is very tricky to get right
- Expensive to collect human rewards at scale
- Human preferences are fallible; models of human preferences even more so

